

STRUCTURE-ONLY FAKE-NEWS DETECTION

Detecting Fake News Without Words

Can topic-agnostic structural sequences match word-based detection?

-
- Viona Widodo
 - Dev Dani
 - Yuto Sugisawa
 - Abigail Zetho Kolondam

Where This Fits

RELATED WORK — AND WHAT THIS PROJECT BORROWS

Prior work	What they did	Shared with this project
Balwant (2019) [1] POS-BiLSTM + CNN · ICCCNT, IEEE · LIAR	• POS-tag sequences read by a BiLSTM • Speaker profile added via a CNN • Structure carries fake-news signal	• Same backbone • Drops the profile metadata • Adds DEP + SHAPE channels • Removes every word
Ahmed, Traore & Saad (2017) [2] n-gram + TF-IDF · ISDDC, Springer	• Word & character n-grams + TF-IDF • Six classical ML classifiers • TF-IDF + Linear SVM $\approx$ 92% accuracy	• Same benchmark • Defines the lexical TF-IDF model • Rebuilt here as the bar to match



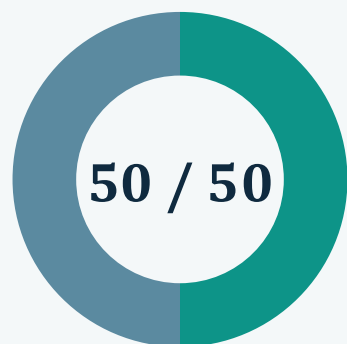
WHY IT INSPIRED THIS PROJECT

Each proved half the case — structure carries signal, words score high. **This project strips the words and asks whether structure alone can match them.**

The Dataset

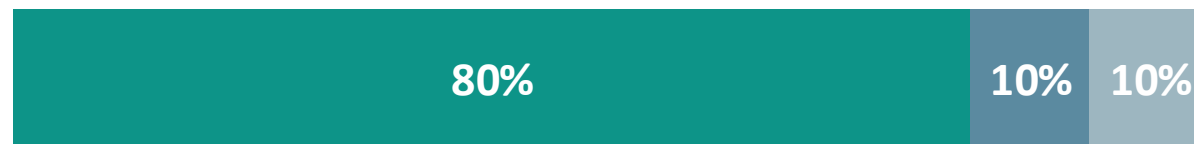
WELFAKE [3] · BALANCED 10,000-ARTICLE SUBSET

Class balance



5,000 fake articles
5,000 real articles

Stratified 80 / 10 / 10 split



● TRAIN

8,000

● VAL

1,000

● TEST

1,000



Fixed split (seed 404), identical across every model & baseline. So all reported numbers are directly comparable, and every significance test is paired on the same 1,000 test items.

From Article to Structure

SPACY [4] PARSE · EVERY ACTUAL WORD DISCARDED

Each article (title + body) is parsed with spaCy and reduced to three parallel token sequences. **All lexical content is removed — only structure remains.**

● **SHAPE** surface form ● **POS** syntax ● **DEP** grammatical relations

token (removed)	Scientists	discovered	a	new	planet	.
SHAPE	Xxxxxx	xxxxx	x	xxx	xxxxx	.
POS	NOUN	VERB	DET	ADJ	NOUN	PUNCT
DEP	nsubj	ROOT	det	amod	dobj	punct

Sequences truncated to 160 tokens. The three channels are topic-agnostic — no vocabulary survives, so nothing ties the model to a subject.

The Model

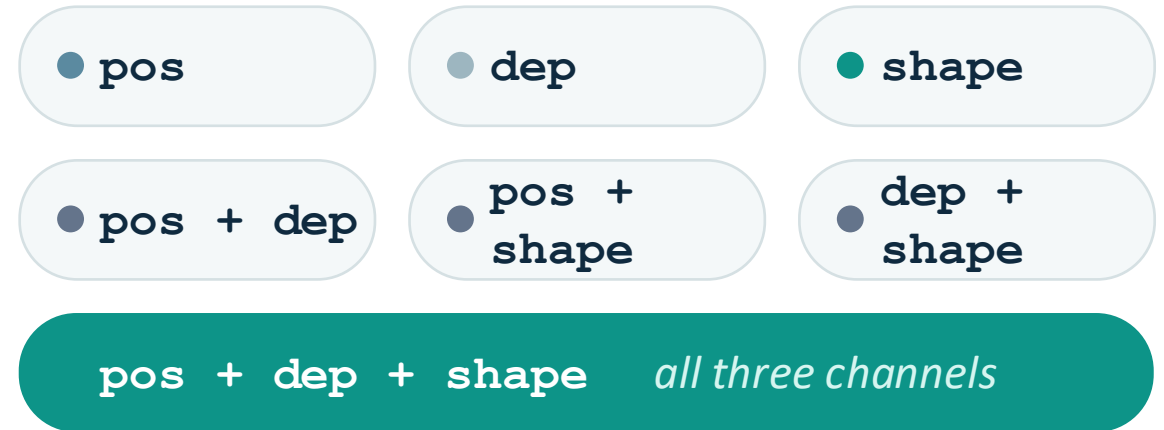
BILSTM OVER STRUCTURAL CHANNELS



Training configuration

Embedding dim	48
Classifier hidden	64
Dropout	0.3
Optimizer	Adam · lr 1e-3
Epochs	6 · early stop on val-F1
Max sequence	160 tokens

Seven channel combinations trained



Baselines & Scoring

WHAT WE COMPARE AGAINST · HOW WE MEASURE

Three reference points



Majority class

FLOOR

Predicts the dominant label — the trivial floor.



Length-only LR

CONTROL

Logistic regression on token count. Tests whether structure is just article length.



Lexical TF-IDF + LR

TARGET

1–2 grams, 50k features on raw words. The content bar the structural models must match.

METRICS

- Accuracy
- Macro precision / recall / F1
- Positive-class F1
- ROC-AUC

Significance McNemar's exact test ($\alpha = 0.05$), paired on the identical test items.

Main Results

TEST SET • SORTED BY MACRO-F1

Model	Accuracy	Macro-F1	ROC-AUC	McNemar vs best
pos_dep_shape (POS+DEP+SHAPE)	0.945	0.945	0.990	reference
pos_shape (POS+SHAPE)	0.941	0.941	0.984	0.63 n.s.
lexical TF-IDF + LR	0.941	0.941	0.989	0.76 n.s.
dep_shape (DEP+SHAPE)	0.930	0.930	0.978	0.086 n.s.
shape (SHAPE only)	0.914	0.914	0.965	0.0004 *
pos_dep (POS+DEP)	0.890	0.890	0.959	<0.001 *
pos (POS only)	0.849	0.849	0.932	<0.001 *
dep (DEP only)	0.811	0.811	0.904	<0.001 *
length-only baseline	0.557	0.542	0.553	<0.001 *
majority-class baseline	0.500	0.333	—	<0.001 *

● Top tier — statistically tied ($p > 0.05$)

n.s. = not significant * = significant at alpha 0.05

THE HEADLINE

Structure alone matches reading every word

0.945

Structure-only macro-F1

POS + DEP + SHAPE

0.941

Lexical TF-IDF macro-F1

sees every word



p = 0.76

McNemar finds no
significant difference

A classifier with all lexical content stripped away is statistically indistinguishable from a strong word-based model.

Which Structure Matters

SURFACE FORM > SYNTAX > DEPENDENCIES

Single channels (macro-F1)



All three single channels differ significantly — including POS vs DEP ($p = 0.004$).

Two channels saturate — the third is redundant

Channel set	F1	Δ	Significance
SHAPE	0.914	—	baseline channel
SHAPE + POS	0.941	+0.027	significant (p=0.001)
SHAPE + DEP	0.930	+0.016	not sig. (p=0.105)
SHAPE+POS+DEP	0.945	+0.031	no gain (p=0.63)



SHAPE is the backbone. One syntactic channel adds real signal; the second is largely subsumed.



Syntax channels complement each other. POS+DEP (0.890) beats POS (0.849) or DEP (0.811) alone — they just add little once SHAPE is present.

Robustness & Next Steps

WHAT STILL NEEDS CHECKING

Second training seed — best model stays on top

Model	F1 seed 404	F1 seed 0	AUC seed 0
pos_dep_shape	0.945	0.967	0.994
pos_shape	0.941	0.925	0.976
dep_shape	0.930	0.948	0.990
shape	0.914	0.913	0.968
pos_dep	0.890	0.875	0.952
pos	0.849	0.825	0.904
dep	0.811	0.825	0.918

Full five-seed mean $\pm$ std sweep still planned.

Open limitations



Single seed

Mid-ranking shifts between seeds; full sweep pending.



Single dataset

All eval within WELFake — cross-dataset test (ISOT / LIAR) needed.



Surface-form signal

May reflect dataset formatting habits; not a length artifact.

Member Contributions

EQUAL SPLIT — 25% EACH

1

Viona Widodo

Data cleaning, feature extraction, and presentation slides and notes

2

Dev Dani

Building the BiLSTM model, presentation recording

3

Yuto Sugisawa

Model training, survey

4

Abigail Zetho Kolondam

Evaluation and ablation study

All four members contributed equally — an even 25% share of the overall project work.

References

1

M. K. Balwant, “Bidirectional LSTM Based on POS Tags and CNN Architecture for Fake News Detection,” in Proc. 10th Int. Conf. Computing, Communication and Networking Technologies (ICCCNT), IEEE, 2019, pp. 1–6.

2

H. Ahmed, I. Traore, and S. Saad, “Detection of Online Fake News Using N-Gram Analysis and Machine Learning Techniques,” in Intelligent, Secure, and Dependable Systems in Distributed and Cloud Environments (ISDDC), LNCS 10618, Springer, 2017, pp. 127–138.

3

P. K. Verma, P. Agrawal, I. Amorim, and R. Prodan, “WELFake: Word Embedding Over Linguistic Features for Fake News Detection,” IEEE Trans. Computational Social Systems, vol. 8, no. 4, pp. 881–893, 2021.

4

M. Honnibal, I. Montani, S. Van Landeghem, and A. Boyd, “spaCy: Industrial-strength Natural Language Processing in Python,” 2020, doi: 10.5281/zenodo.1212303.